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The Editors
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Dear Editors,

I am pleased to submit for your consideration the enclosed manuscript, **"Investment Gestation Lags, Intangible Capital, and the Measurement of Capital and Productivity in 39 Economies"**, for publication in the *Oxford Bulletin of Economics and Statistics*.

The paper addresses a practical measurement problem in applied growth and productivity analysis. Standard national-accounts capital construction treats investment as productive the moment it is spent and omits much of intangible investment from produced capital — two conventions that were defensible under the old investment mix and are increasingly hard to defend now that capital formation has shifted toward software, R&D platforms, and other long-gestation assets. Because total factor productivity is computed as a residual, the resulting timing and coverage errors are absorbed into measured productivity and re-interpreted as technical change. I quantify both corrections jointly: a time-varying investment-to-output lag $\mu(t)$ and a country-specific intangible-capital share β, estimated for 39 OECD and middle-income economies over 1970–2019, with the production and wealth accounts required to agree on the same parameters.

The manuscript sits directly in the line of work the *Bulletin* has published on intangible capital and productivity measurement — dal Borgo, Goodridge, Haskel and Pesole's (2013) industry-level treatment of intangibles as investment, and Corrado, Haskel and Jona-Lasinio's (2017) analysis of intangible-capital spillovers across EU economies. Where those papers expand the *definition* of investment, this paper adds the missing *timing* dimension: treating the two omissions jointly shows that part of what measured capital — and hence the Solow residual — records is a gestation-lag artefact rather than genuine input or technical change.

The headline results are deliberately modest and reported with their uncertainty. Correcting the gestation-lag convention lowers measured capital by a median 4.3 per cent across the 35 countries with asset-type detail, raising log-level TFP and the implied labour share by a median 1.7 percentage points each. The joint production–wealth identification survives a strict temporal holdout on the wealth series: parameters fitted on 1995–2014 produce a smaller out-of-sample residual than the production-only alternative in 23 of 39 countries. The corresponding improvement in out-of-sample level accuracy (median MAPE 4.60 % to 3.99 %) is reported together with its non-significant Wilcoxon test, and bootstrap intervals are presented as interval rather than point identification. A parameter-free observable proxy built from OECD asset-composition data reproduces the direction of the fitted corrections, so the core adjustment can be replicated at negligible cost by any statistical agency or applied researcher.

I believe the paper suits the *Bulletin*'s readership precisely because the contribution is practical: it changes how applied economists should read capital stocks, TFP levels, labour shares, and flow–stock discrepancies in the public datasets the profession uses every day — PWT, CWON, WDI, and OECD GFCF — rather than proposing a new theoretical mechanism.

I confirm the following:

- This work is original and has not been published elsewhere.
- The manuscript is not under consideration by any other journal.
- All data sources (Penn World Table 10.01; World Bank Changing Wealth of Nations; World Bank WDI; OECD GFCF) are publicly available; frozen extracts, code, and intermediate outputs are archived at https://github.com/bougtoir/gdp-tempo-paper, and a one-command pipeline regenerates every figure, table, and number in the manuscript.
- I am the sole author.
- I have no competing interests, financial or otherwise, related to this work.
- I used generative AI to assist with formatting the text, choosing words, and writing analysis code; this is documented in the Declarations section of the title page.

A title page with full author details and a Declarations section is included as a separate file.

I would be grateful for your consideration.

Yours sincerely,

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